Public: Module 01: Chapter 01: Balance Sheets

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Cash & **Cash Equivalents**
+Short-Term Investments
+Accounts & Other Receivables
+Inventories
+Other Current Assets
=**Total Current Assets** (**TCA**)
+Property, Plant & Equipment (PP&E, Net)
+Goodwill
+Intangible Assets (Net)
+Other Non-Current Assets
=**Total Assets** (**TA**)

Following the common-size balance sheet framework presented by BARUCH-FIN3000 [2019].

1 Balance Sheet: Total Assets

Across a sample universe of approximately 300 corporate entities spanning a 10-year period (9,600+ quarterly balance sheet records), reported top-level metrics exhibit near-perfect baseline integrity. This serves as good common scaler in both mathematical properties and econometrical properties Cooper et al. [2008].

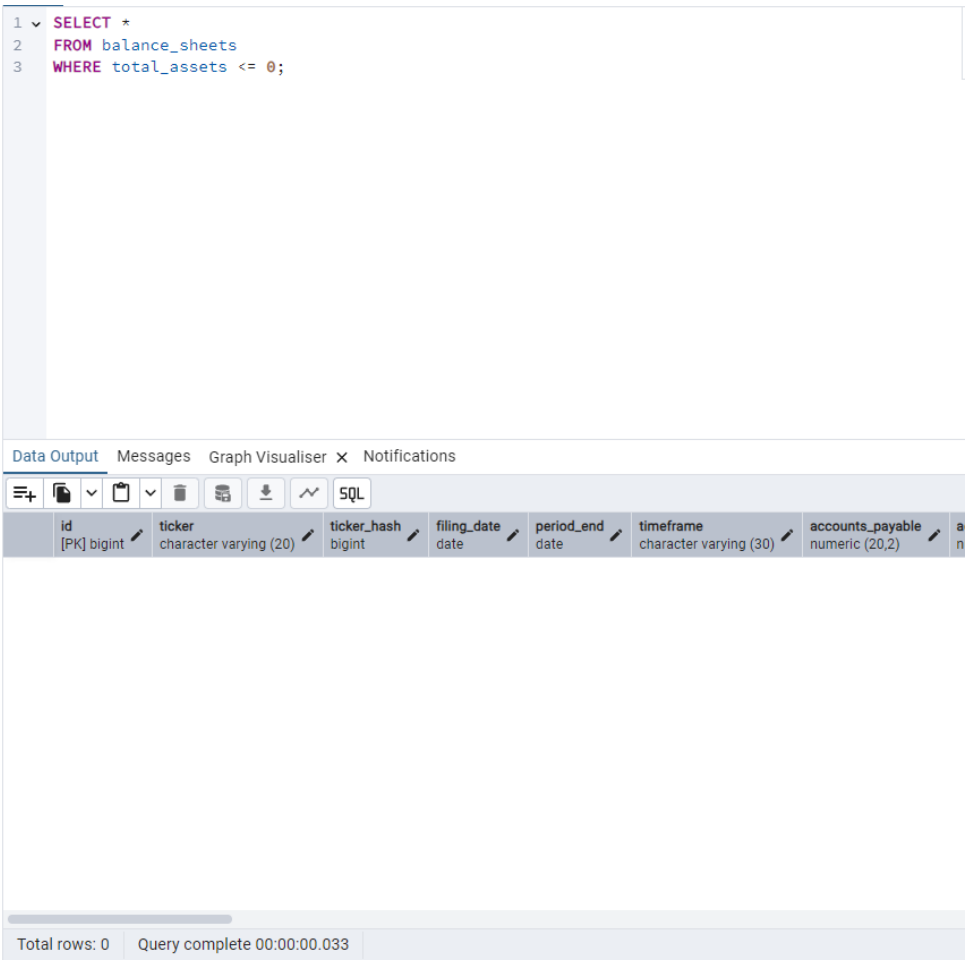


Figure 1: This validation returns exactly zero records, confirming that reported total_assets is reliably positive across all non-null filings.

1.1 Deduplication on Latest Filing

Multiple balance sheet observations frequently exist for a single (ticker, period_end) pair due to subsequent restatements, 10-K/A amendments, and preliminary earnings releases. Deduplicating by canonical ingestion order requires a partitioned window ranking by descending filing_date.

Listing 1 demonstrates the reconciliation check between annual and quarterly filings at identical period-ends.

```sql
WITH latest_balance_sheets AS (
    SELECT
        *,
        ROW_NUMBER() OVER (
            PARTITION BY ticker, period_end, timeframe
            ORDER BY filing_date DESC
        ) AS rn
    FROM balance_sheets
)
SELECT
    a.ticker,
    a.period_end,
    a.total_assets AS annual_assets,
    q.total_assets AS quarterly_assets,
    (a.total_assets - q.total_assets) AS diff
FROM latest_balance_sheets a
INNER JOIN latest_balance_sheets q
    ON  a.ticker = q.ticker
    AND a.period_end = q.period_end
    AND q.timeframe = 'quarterly'
    AND q.rn = 1
WHERE a.timeframe = 'annual'
  AND a.rn = 1
ORDER BY diff DESC;
```

Listing 1: Annual vs. Quarterly Balance Sheet Reconciliation via Window Deduplication

Query | Query History

```sql
WITH latest_balance_sheets AS (
    SELECT
        *,
        ROW_NUMBER() OVER (
            PARTITION BY ticker, period_end, timeframe
            ORDER BY filing_date DESC
        ) AS rn
    FROM balance_sheets
)
SELECT
    a.ticker,
    a.period_end,
    a.total_assets AS annual_assets,
    q.total_assets AS quarterly_assets,
    (a.total_assets - q.total_assets) AS diff
FROM latest_balance_sheets a
INNER JOIN latest_balance_sheets q
    ON  a.ticker = q.ticker
    AND a.period_end = q.period_end
    AND q.timeframe = 'quarterly'
    AND q.rn = 1
WHERE a.timeframe = 'annual'
  AND a.rn = 1
ORDER BY diff DESC;
```

Data Output | Messages | Graph Visualiser | Notifications

	ticker character varying (20)	period_end date	annual_assets numeric (20,2)	quarterly_assets numeric (20,2)	diff numeric
1	A	2016-10-31	7794000000.00	7794000000.00	0.00
2	A	2017-10-31	8426000000.00	8426000000.00	0.00
3	A	2018-10-31	8541000000.00	8541000000.00	0.00
4	A	2019-10-31	9452000000.00	9452000000.00	0.00
5	A	2020-10-31	9627000000.00	9627000000.00	0.00
6	A	2021-10-31	10705000000.00	10705000000.00	0.00
7	A	2022-10-31	10532000000.00	10532000000.00	0.00
8	A	2023-10-31	10763000000.00	10763000000.00	0.00

Figure 2: Reconciliation with latest-filing deduplication (diff = 0).

```sql
SELECT
    a.ticker,
    a.period_end,
    a.total_assets AS annual_assets,
    q.total_assets AS quarterly_assets,
    (a.total_assets - q.total_assets) AS diff
FROM balance_sheets a
INNER JOIN balance_sheets q
    ON  a.ticker = q.ticker
    AND a.period_end = q.period_end
    AND q.timeframe = 'quarterly'
WHERE a.timeframe = 'annual'
ORDER BY diff DESC;
```

Data Output | Messages | Graph Visualiser | Notifications

	ticker character varying (20)	period_end date	annual_assets numeric (20,2)	quarterly_assets numeric (20,2)	diff numeric
1	EP	2025-12-31	72748000000.00	65873000.00	72682127000.00
2	EP	2024-12-31	71407000000.00	123868000.00	71283132000.00
3	EP	2023-12-31	71020000000.00	92615616.00	70927384384.00
4	EP	2022-12-31	70078000000.00	71545075.00	70006454925.00
5	KO	2024-12-31	100549000000.00	99392000000.00	1157000000.00
6	TXN	2021-12-31	24676000000.00	24676000000.00	0.00
7	TXN	2022-12-31	27207000000.00	27207000000.00	0.00
8	TXN	2023-12-31	32348000000.00	32348000000.00	0.00
9	TXN	2024-12-31	35509000000.00	35509000000.00	0.00

Total rows: 2444 Query complete 00:00:00.051

Figure 3: Severe variance induced by non-canonical or arbitrary record ordering.

Figure 4: Impact of filing-date deduplication policies on balance sheet reconciliation.

Query Query History

```sql
WITH latest_annuals AS (
    SELECT
        *,
        ROW_NUMBER() OVER (
            PARTITION BY ticker, period_end
            ORDER BY filing_date DESC
        ) AS rn
    FROM balance_sheets
    WHERE timeframe = 'annual'
)
SELECT
    a.ticker,
    a.period_end,
    a.filing_date,
    a.total_assets
FROM latest_annuals a
WHERE a.rn = 1
  AND NOT EXISTS (
      SELECT 1
      FROM balance_sheets q
      WHERE q.ticker = a.ticker
        AND q.period_end = a.period_end
        AND q.timeframe = 'quarterly'
  )
ORDER BY a.ticker ASC, a.period_end DESC;
```

Data Output Messages Graph Visualiser Notifications

SQL

ticker character varying (20)	period_end date	filing_date date	total_assets numeric (20,2)

Total rows: 0 Query complete 00:00:00.218

Figure 5: Verification showing complete quarterly representation for every annual filing period.

1.2 Current Asset Mapping Deficiencies

While reported aggregates (total_current_assets, total_assets) remain reliable audited anchors, reconstructing these aggregates from standard reporting line items surfaces parser classification flaws.

Out of 9,600 quarterly entries, only a single manufacturing line-item parsing omission was detected: Northrop Grumman (NOC, period ended September 30, 2023), where an unmapped XBRL cash tag caused an omitted $2.082B line item against the correctly reported total_current_assets total (Figure 6).

Query Query History

```
    bs.ticker,
    bs.period_end,
    bs.cash_and_equivalents,
    bs.short_term_investments,
    bs.receivables,
    bs.inventories,
    bs.other_current_assets,
    bs.total_current_assets as tca,
    (COALESCE(bs.cash_and_equivalents, 0)
    + COALESCE(bs.short_term_investments, 0)
    + COALESCE(bs.receivables, 0)
    + COALESCE(bs.inventories, 0)
    + COALESCE(bs.other_current_assets, 0)
    )  as r_tca,
    (COALESCE(bs.cash_and_equivalents, 0)
    + COALESCE(bs.short_term_investments, 0)
    + COALESCE(bs.receivables, 0)
    + COALESCE(bs.inventories, 0)
    + COALESCE(bs.other_current_assets, 0)
    ) - bs.total_current_assets AS diff



FROM latest_balance_sheets bs
WHERE bs.timeframe = 'quarterly'
  AND bs.rn = 1
    AND (
    (COALESCE(bs.cash_and_equivalents, 0)
    + COALESCE(bs.short_term_investments, 0)
    + COALESCE(bs.receivables, 0)
    + COALESCE(bs.inventories, 0)
    + COALESCE(bs.other_current_assets, 0)
    ) - bs.total_current_assets) <> 0
ORDER BY diff DESC;
```

Data Output Messages Graph Visualiser Notifications

Showing rows: 1 to 1 Page No: 1

	ticker character varying (20)	period_end date	cash_and_equivalents numeric (20,2)	short_term_investments numeric (20,2)	receivables numeric (20,2)	inventories numeric (20,2)	other_current_assets numeric (20,2)	tca numeric (20,2)	r_tca numeric	diff numeric
1	NOC	2023-09-30	[null]	0.00	8034000000.00	1355000000.00	1679000000.00	13150000000.00	11068000000.00	-2082000000.00

Total rows: 1 Query complete 00:00:00.046

Figure 6: Northrop Grumman (NOC) line-item ingestion omission in Q3 2023.

1.3 Total Asset Component Reconciliation

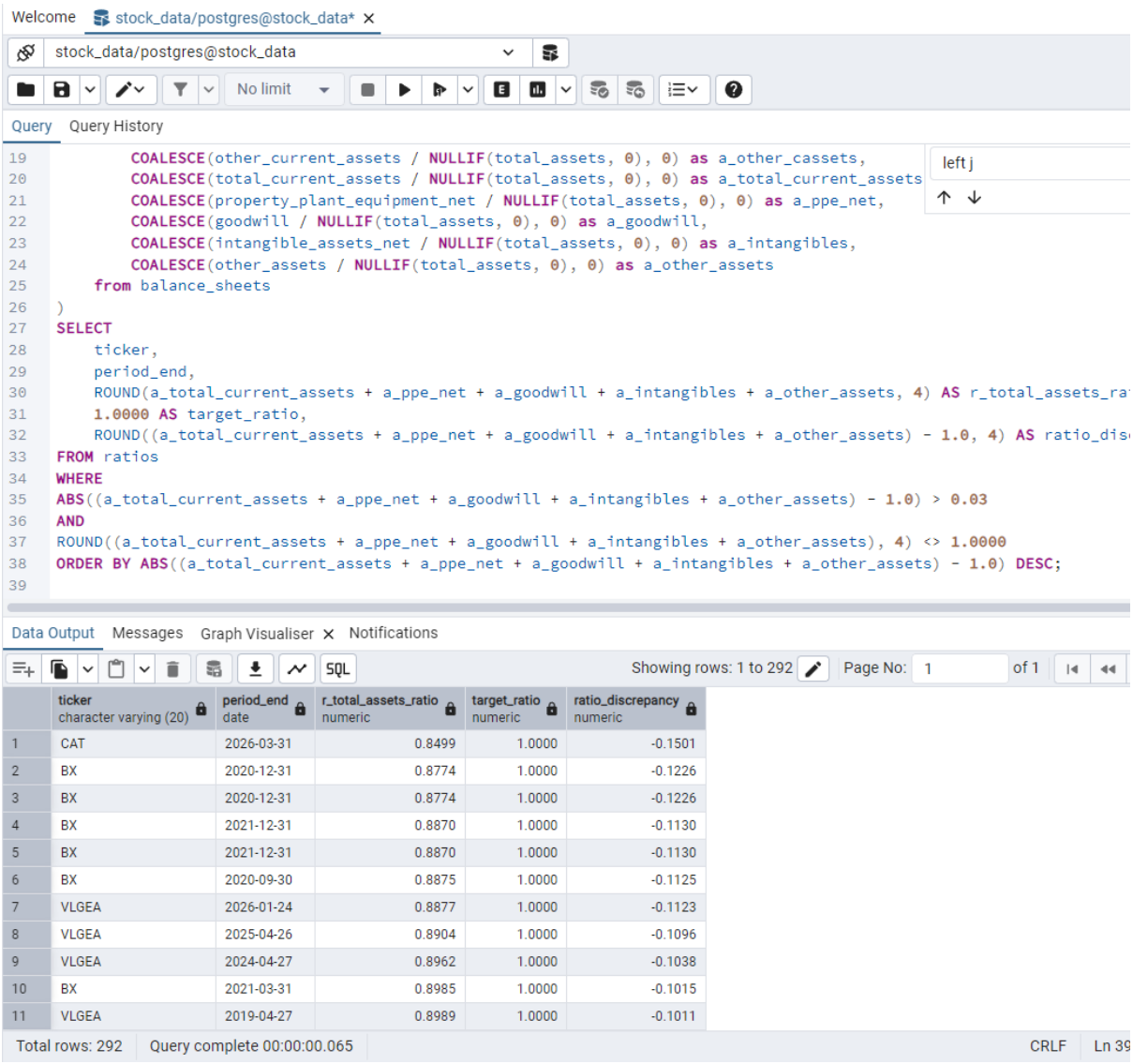
Listing 2 evaluates the discrepancy between reported total assets and reconstructed assets:

$$\text{Asset Coverage} = \frac{\text{Current Assets} + \text{Net PP\&E} + \text{Goodwill} + \text{Net Intangibles} + \text{Other Assets}}{\text{Reported Total Assets}} \tag{1}$$

```sql

WITH latest_balance_sheets AS (
    SELECT
        *,
        ROW_NUMBER() OVER (
            PARTITION BY ticker, period_end, timeframe
            ORDER BY filing_date DESC
        ) AS rn
    FROM balance_sheets
),
ratios as (
    select
        ticker,
        period_end,
        timeframe,
        COALESCE(cash_and_equivalents / NULLIF(total_assets, 0), 0) as a_cash,
        COALESCE(short_term_investments / NULLIF(total_assets, 0), 0) as a_short_invest,
        COALESCE(receivables / NULLIF(total_assets, 0), 0) as a_receiv,
        COALESCE(inventories / NULLIF(total_assets, 0), 0) as a_invent,
        COALESCE(other_current_assets / NULLIF(total_assets, 0), 0) as a_other_cassets,
        COALESCE(total_current_assets / NULLIF(total_assets, 0), 0) as
    a_total_current_assets,
        COALESCE(property_plant_equipment_net / NULLIF(total_assets, 0), 0) as a_ppe_net
    ,
        COALESCE(goodwill / NULLIF(total_assets, 0), 0) as a_goodwill,
        COALESCE(intangible_assets_net / NULLIF(total_assets, 0), 0) as a_intangibles,
        COALESCE(other_assets / NULLIF(total_assets, 0), 0) as a_other_assets
    from latest_balance_sheets
)
SELECT
    ticker,
    period_end,
    ROUND(a_total_current_assets + a_ppe_net + a_goodwill + a_intangibles +
    a_other_assets, 4) AS r_total_assets_ratio,
    1.0000 AS target_ratio,
    ROUND((a_total_current_assets + a_ppe_net + a_goodwill + a_intangibles +
    a_other_assets) - 1.0, 4) AS ratio_discrepancy
FROM ratios
  WHERE
  ABS((a_total_current_assets + a_ppe_net + a_goodwill + a_intangibles + a_other_assets)
     - 1.0) > 0.03
  AND ROUND((a_total_current_assets + a_ppe_net + a_goodwill + a_intangibles +
    a_other_assets), 4) <> 1.0000
ORDER BY ABS((a_total_current_assets + a_ppe_net + a_goodwill + a_intangibles +
    a_other_assets) - 1.0) DESC;
```

Listing 2: Total Asset Reconstruction Discrepancy Query



```
        COALESCE(other_current_assets / NULLIF(total_assets, 0), 0) as a_other_cassets,
        COALESCE(total_current_assets / NULLIF(total_assets, 0), 0) as a_total_current_assets
        COALESCE(property_plant_equipment_net / NULLIF(total_assets, 0), 0) as a_ppe_net,
        COALESCE(goodwill / NULLIF(total_assets, 0), 0) as a_goodwill,
        COALESCE(intangible_assets_net / NULLIF(total_assets, 0), 0) as a_intangibles,
        COALESCE(other_assets / NULLIF(total_assets, 0), 0) as a_other_assets
    from balance_sheets
)
SELECT
    ticker,
    period_end,
    ROUND(a_total_current_assets + a_ppe_net + a_goodwill + a_intangibles + a_other_assets, 4) AS r_total_assets_ra
    1.0000 AS target_ratio,
    ROUND((a_total_current_assets + a_ppe_net + a_goodwill + a_intangibles + a_other_assets) - 1.0, 4) AS ratio_dis
FROM ratios
WHERE
ABS((a_total_current_assets + a_ppe_net + a_goodwill + a_intangibles + a_other_assets) - 1.0) > 0.03
AND
ROUND((a_total_current_assets + a_ppe_net + a_goodwill + a_intangibles + a_other_assets), 4) <> 1.0000
ORDER BY ABS((a_total_current_assets + a_ppe_net + a_goodwill + a_intangibles + a_other_assets) - 1.0) DESC;
```

	ticker character varying (20)	period_end date	r_total_assets_ratio numeric	target_ratio numeric	ratio_discrepancy numeric
1	CAT	2026-03-31	0.8499	1.0000	-0.1501
2	BX	2020-12-31	0.8774	1.0000	-0.1226
3	BX	2020-12-31	0.8774	1.0000	-0.1226
4	BX	2021-12-31	0.8870	1.0000	-0.1130
5	BX	2021-12-31	0.8870	1.0000	-0.1130
6	BX	2020-09-30	0.8875	1.0000	-0.1125
7	VLGEA	2026-01-24	0.8877	1.0000	-0.1123
8	VLGEA	2025-04-26	0.8904	1.0000	-0.1096
9	VLGEA	2024-04-27	0.8962	1.0000	-0.1038
10	BX	2021-03-31	0.8985	1.0000	-0.1015
11	VLGEA	2019-04-27	0.8989	1.0000	-0.1011

Figure 7: Distribution of 292 rows exhibiting reconstruction variance exceeding a 3% tolerance threshold, primarily driven by financial institutions, captive finance loans, and goodwill nesting.

1.4 Total Assets Unit Scaling Anomaly

1.4.1 Theoretical Framework: Unit Discontinuities vs. Economic Innovations

Cross-statement accounting identities normalize error terms by total assets (Error/Assets) to ensure cross-sectional comparability. However, this normalization implicitly presumes that the scaling denominator itself possesses stationary numerical fidelity. Under SEC EDGAR XBRL ingestion, data feeds frequently drop or misapply unit multiplier tags (e.g., misinterpreting an integer reported in thousands, $000, as whole base dollars).

To detect and isolate unit scale corruption from real corporate transformations, the balance sheet time series must be formulated as a dynamical innovation process via the quarterly log return of total assets:

$$r_{\text{Assets},t} = \ln\left(\frac{\text{Assets}_t}{\text{Assets}_{t-1}}\right) \tag{2}$$

Under ordinary macroeconomic conditions, organic quarterly asset growth is bounded within $|r_{\text{Assets},t}| \leq 0.50$. In contrast, an unscaled thousands-to-dollars unit ingestion omission produces a discrete, reversible step discontinuity of exact magnitude:

$$|\Delta r_{\text{Unit}}| = \ln(10^3) = 3\ln(10) \approx \mathbf{6.907755} \tag{3}$$

Because genuine economic growth cannot expand or contract an enterprise's physical asset footprint by 1,000× in a single 90-day calendar interval and immediately invert in the subsequent quarter, symmetric innovation pairs where $r_t \approx -6.91$ and $r_{t+1} \approx +6.91$ serve as an exact identifier of unit-order parsing corruption.

Listing 3 provides the production SQL implementation tracking quarterly log innovations across canonical balance sheets.

```sql
WITH bs_raw AS (
    -- 1. Deduplicate by latest filing per distinct filing period and timeframe
    SELECT
        ticker,
        ticker_hash,
        fiscal_year,
        fiscal_quarter,
        timeframe,
        period_end,
        total_assets,
        ROW_NUMBER() OVER (
            PARTITION BY ticker_hash, fiscal_year, fiscal_quarter, timeframe
            ORDER BY filing_date DESC
        ) AS rn
    FROM balance_sheets
    WHERE timeframe IN ('quarterly', 'annual')
      AND total_assets IS NOT NULL
      AND total_assets > 0
),
bs_canonical AS (
    -- 2. Leave Q1-Q3 alone (quarterly). For Q4, deduplicate selecting 'annual' over '
    quarterly'
    SELECT
        ticker,
        ticker_hash,
        fiscal_year,
        fiscal_quarter,
        period_end,
        total_assets,
        ROW_NUMBER() OVER (
            PARTITION BY ticker_hash, fiscal_year, fiscal_quarter
            ORDER BY
                CASE
                    WHEN fiscal_quarter = 4 AND timeframe = 'annual' THEN 1
                    WHEN fiscal_quarter != 4 AND timeframe = 'quarterly' THEN 1
                    ELSE 2
                END,
                period_end DESC
```

```sql
        ) AS rn_period
    FROM bs_raw
    WHERE rn = 1
      AND (
          (fiscal_quarter IN (1, 2, 3) AND timeframe = 'quarterly')
          OR
          (fiscal_quarter = 4)
      )
),
bs_deduped AS (
    SELECT
        ticker,
        ticker_hash,
        fiscal_year,
        fiscal_quarter,
        period_end,
        total_assets
    FROM bs_canonical
    WHERE rn_period = 1
),
bs_sequenced AS (
    -- 3. Chronological lag strictly by period_end to link consecutive quarters
    SELECT
        ticker,
        ticker_hash,
        fiscal_year,
        fiscal_quarter,
        period_end AS curr_period_end,
        total_assets AS curr_total_assets,

        LAG(period_end) OVER (
            PARTITION BY ticker_hash
            ORDER BY period_end ASC
        ) AS prev_period_end,

        LAG(fiscal_quarter) OVER (
            PARTITION BY ticker_hash
            ORDER BY period_end ASC
        ) AS prev_fiscal_quarter,

        LAG(total_assets) OVER (
            PARTITION BY ticker_hash
            ORDER BY period_end ASC
        ) AS prev_total_assets
    FROM bs_deduped
),
asset_returns AS (
    -- 4. Materialize deltas and log returns
    SELECT
        ticker,
        prev_period_end,
        prev_fiscal_quarter,
        curr_period_end,
        fiscal_quarter AS curr_fiscal_quarter,
        prev_total_assets,
        curr_total_assets,
        ROUND((curr_total_assets - prev_total_assets), 2) AS total_assets_diff,
        ROUND(LN(curr_total_assets / prev_total_assets), 6) AS log_return_total_assets
    FROM bs_sequenced
    WHERE prev_total_assets IS NOT NULL
      AND prev_total_assets > 0
      AND curr_total_assets > 0
      -- Guardrail: enforce consecutive quarter calendar adjacency (allows 60 to 105
    days)
      AND (curr_period_end - prev_period_end BETWEEN 60 AND 105)
)
SELECT *
FROM asset_returns
ORDER BY ABS(log_return_total_assets) DESC;
```

Listing 3: Sequential Total Assets Log-Return Audit for Unit Multiplier and Structural Shock Detection

Query Query History

```
81          ) AS prev_total_assets
82      FROM bs_deduped
83  ),
84  asset_returns AS (
85      -- 4. Materialize deltas and log returns
86      SELECT
87          ticker,
88          prev_period_end,
89          prev_fiscal_quarter,
90          curr_period_end,
91          fiscal_quarter AS curr_fiscal_quarter,
92          prev_total_assets,
93          curr_total_assets,
94          ROUND((curr_total_assets - prev_total_assets), 2) AS total_assets_diff,
95          ROUND(LN(curr_total_assets / prev_total_assets), 6) AS log_return_total_assets
96      FROM bs_sequenced
97      WHERE prev_total_assets IS NOT NULL
98        AND prev_total_assets > 0
99        AND curr_total_assets > 0
100       -- Guardrail: enforce consecutive quarter calendar adjacency (allows 60 to 105 days)
101       AND (curr_period_end - prev_period_end BETWEEN 60 AND 105)
102 )
103 SELECT *
104 FROM asset_returns
105 ORDER BY ABS(log_return_total_assets) DESC;
```

Data Output Messages Notifications

Showing rows: 1 to 1

	ticker character varying (20)	prev_period_end date	prev_fiscal_quarter integer	curr_period_end date	curr_fiscal_quarter integer	prev_total_assets numeric	curr_total_assets numeric (20,2)	total_assets_diff numeric	log_return_total_assets numeric
1	COF	2019-06-30	2	2019-09-30	3	373619000.00	378810000000.00	378436381000.00	6.921553
2	COF	2019-03-31	1	2019-06-30	2	373191000000.00	373619000.00	-372817381000.00	-6.906609
3	COIN	2023-12-31	4	2024-03-31	1	14753901000.00	348036477000.00	333282576000.00	3.160800
4	CNTX	2021-09-30	3	2021-12-31	4	2259304.00	51305750.00	49046446.00	3.122746
5	COIN	2024-09-30	3	2024-12-31	4	290558608000.00	22541951000.00	-268016657000.00	-2.556427
6	CBOE	2016-12-31	4	2017-03-31	1	476700000.00	5345100000.00	4868400000.00	2.417048
7	ASST	2024-09-30	3	2024-12-31	4	2553092.00	28197000.00	25643908.00	2.401910
8	FCUV	2025-09-30	3	2025-12-31	4	880445.00	8885514.00	8005069.00	2.311750
9	CNTX	2024-03-31	1	2024-06-30	2	11051652.00	102708853.00	91657201.00	2.229318
10	COIN	2023-09-30	3	2023-12-31	4	127305595000.00	14753901000.00	-112551694000.00	-2.155083
11	TIPT	2026-03-31	1	2026-06-30	2	6967655000.00	1134117000.00	-5833538000.00	-1.815424
12	ASST	2024-12-31	4	2025-03-31	1	28197000.00	4994288.00	-23202712.00	-1.730921

Total rows: 9100 Query complete 00:00:00.110 Rows selected: 15

Figure 8: Empirical distribution of consecutive quarterly total asset log returns across the canonical corporate filing panel: extreme tail innovations cleanly segment raw ingestion unit omission anomalies ($|r| \approx 6.91$) from macro-regulatory shifts and major corporate acquisitions ($|r| \in [1.5, 3.5]$).

1.4.2 Empirical Evaluation and Outlier Patterns

Executing Listing 3 isolates the highest-variance innovations across the database, compiled in Table 1. Rather than representing homogenous pipeline errors, these observations fall into structural clusters of parsing faults, accounting rule transitions, and major corporate events.

Table 1: Quarterly Total Assets Log-Return Innovation Outliers

Ticker	Prev End	FQ_0	Curr End	FQ_1	Prev Assets ($M)	Curr Assets ($M)	ΔAssets ($M)	Log Return	Structural Classification
COF	2019-06-30	2	2019-09-30	3	373.62	378,810.00	378,436.38	**6.9216**	Unit Recovery ($\times 10^3$)
COF	2019-03-31	1	2019-06-30	2	373,191.00	373.62	-372,817.38	**-6.9066**	Unit Omission ($\div 10^3$)
COIN	2023-12-31	4	2024-03-31	1	14,753.90	348,036.48	333,282.58	3.1608	Regulatory Influx (SAB 121)
CNTX	2021-09-30	3	2021-12-31	4	2.26	51.31	49.05	3.1227	Clinical Reverse Merger Recap
COIN	2024-09-30	3	2024-12-31	4	290,558.61	22,541.95	-268,016.66	-2.5564	Custodial Liability Carve-out
CBOE	2016-12-31	4	2017-03-31	1	476.70	5,345.10	4,868.40	2.4170	M&A (Bats Global Acquisition)
ASST	2024-09-30	3	2024-12-31	4	2.55	28.20	25.64	2.4019	Registered Direct Offering
FCUV	2025-09-30	3	2025-12-31	4	0.88	8.89	8.01	2.3118	Reverse Split Capital Influx
CNTX	2024-03-31	1	2024-06-30	2	11.05	102.71	91.66	2.2293	Pipe Financing Equity Injection
COIN	2023-09-30	3	2023-12-31	4	127,305.60	14,753.90	-112,551.70	-2.1551	Custodial Reclassification
TIPT	2026-03-31	1	2026-06-30	2	6,967.66	1,134.12	-5,833.54	-1.8154	Insurance Carve-Out Divestiture
ASST	2024-12-31	4	2025-03-31	1	28.20	4.99	-23.20	-1.7309	Operating Cash Depletion
NDRA	2018-09-30	3	2018-12-31	4	1.29	7.22	5.93	1.7211	Convertible Note Liquidity
AMD	2021-12-25	4	2022-03-26	1	12,419.00	66,915.00	54,496.00	1.6842	M&A (Xilinx Mega-Acquisition)
CNSP	2020-09-30	3	2020-12-31	4	2.99	15.85	12.87	1.6697	Equity Line Consolidation

1.4.3 Root Cause Taxonomy

1. **Unit Ingestion Multiplier Discontinuities ($|r| \approx 6.91$):** Capital One Financial Corporation (`COF`) presents the textbook signature of a missing unit scale tag. Assets contract from \$373.19B in 2019-Q1 to \$373.62M in 2019-Q2 ($r = -6.9066$), followed immediately by an expansion to \$378.81B in 2019-Q3 ($r = +6.9216$). The exact symmetry around $\pm \ln(10^3)$ confirms that the Q2 Form 10-Q parser failed to register the `thousands` reporting scale, recording unadjusted thousands as whole dollars.

2. **Regulatory Accounting Presentation Shocks (`COIN`):** Coinbase Global exhibits massive volatility across 2023 and 2024 ($+3.1608$, -2.5564, -2.1551). These swings stem from SEC Staff Accounting Bulletin No. 121 (SAB 121), which mandated that digital asset custodians recognize customer-safeguarded crypto liabilities on-balance-sheet alongside a matching safeguarding asset. When crypto valuations fluctuated and subsequent legal interpretations allowed carve-outs from balance sheet gross-ups, hundreds of billions in custodial assets entered and exited the balance sheet without operational equity impact.

3. **Strategic Mega-M&A Additions (`AMD`, `CBOE`):** Advanced Micro Devices (`AMD`) expanded its balance sheet by \$54.50B in 2022-Q1 ($r = +1.6842$) upon closing its \$49B acquisition of Xilinx. Similarly, CBOE Global Markets grew by \$4.87B in 2017-Q1 ($r = +2.4170$) upon consummating its purchase of Bats Global Markets. These large innovations reflect recognized goodwill, identifiable intangible assets, and acquired working capital.

4. **Micro-Cap Equity Offerings and Clinical Recapitalizations (`CNTX`, `ASST`, `FCUV`):** Small-cap issuers operating with minimal asset bases ($<$ \$10M) routinely generate $|r| \in [1.5, 3.2]$ through dilutive financing lines, PIPE transactions, or reverse recapitalizations. A \$25M direct registered offering by `ASST` in 2024-Q4 expanded assets tenfold from \$2.55M to \$28.20M ($r = +2.4019$), only to contract back to \$4.99M in 2025-Q1 ($r = -1.7309$) upon operational expenditure and liability retirement.

1.4.4 Automated Pipeline Resolution: The Unit Quenching Layer

To protect cross-statement denominator stability without suppressing authentic corporate M&A transactions or SAB 121 regulatory regimes, automated ETL systems must deploy a multi-period unit-quenching heuristic prior to calculating financial statement ratios:

$$\text{Assets}_{t,\text{Quenched}} = \begin{cases} \text{Assets}_t \times 10^3, & \text{if } |r_t - (-6.908)| \leq 0.15 \text{ and } |r_{t+1} - 6.908| \leq 0.15 \\ \text{Assets}_t \times 10^{-3}, & \text{if } |r_t - 6.908| \leq 0.15 \text{ and } |r_{t+1} - (-6.908)| \leq 0.15 \\ \text{Assets}_t, & \text{otherwise} \end{cases} \quad (4)$$

By asserting sequential reversibility within a tight tolerance band around $\pm \ln(10^3)$, the data pipeline corrects isolated multiplier errors while preserving one-way economic shifts such as the Xilinx acquisition or Coinbase crypto custody accounting.

1.5 Point-in-Time Resolution Error Modes

Don't scale by anything other that 1e-3 or 1e3 base 10. If you have 20k a year and get a secondary data source. Otherwise just pray to god that there are no 1e1 base 10 formatting errors (It exist, ingestion date 08/30/2026 NVDA point in time date 2022, basic shares outstanding understated by 1e10 base 10,l meaning this error affects everybody regardless of cap), and that errors are all just ASC 121-122 and 805 restatements.

1.6 Architectural Premise and Invariant Constraints

Fundamental data ingestion cannot assume cross-statement synchrony or temporal alignment across reporting regimes. To preserve true Point-in-Time (PIT) integrity without introducing lookahead bias or corrupting covariance estimation, the ingestion resolver evaluates candidate records under three foundational invariants:

1. **Accounting Invariant:** Within any self-contained filing F, the balance sheet identity must hold within numerical tolerance:

   $$|\text{Total Assets} - (\text{Total Liabilities} + \text{Total Stockholders' Equity})| \leq \epsilon \cdot \text{Total Assets} \quad (5)$$

   If this condition holds on raw extraction, no scalar multiplier may be applied.

2. **Discrete Scale Quantization:** Vendor parsing failures occur almost exclusively along discrete powers of ten matching SEC reporting units:

   $$s \in \{10^{-6}, 10^{-3}, 1.0, 10^3, 10^6\} \quad (6)$$

   Arbitrary fractional scaling or $10^{\pm 1}$ continuous adjustments are strictly prohibited.

3. **Accession-Date Indexing:** Observations must be strictly ordered by SEC acceptance timestamp τ_{accept} rather than period end t_{period} alone to prevent retrospective restatement contamination.

1.7 Failure Taxonomy and Resolution Mechanics

1.7.1 Cases 1 & 2: Homogeneous Discrete Unit Scaling Glitches

Symptom:

- *Case 1 (Share Count):* $\text{Shares}_{t-1} \approx 400\text{M} \longrightarrow \text{Shares}_t = 400\text{K} \longrightarrow \text{Shares}_{t+1} \approx 400\text{M}$.
- *Case 2 (Total Assets):* $\text{Assets}_{t-1} \approx \$300\text{B} \longrightarrow \text{Assets}_t = \$300\text{M} \longrightarrow \text{Assets}_{t+1} \approx \$300\text{B}$.

Diagnosis: The extraction pipeline omitted a document-level multiplier tag (e.g., *"Dollars/Shares in thousands"*), yielding an exact drop by factor 10^3:

$$\ln\left(\frac{x_{t-1}}{x_t}\right) \approx \ln(1000) \approx 6.908, \quad \ln\left(\frac{x_{t+1}}{x_t}\right) \approx \ln(1000) \approx 6.908 \tag{7}$$

Resolution Rule: Verify cross-statement consistency. If corroborating line items in the same filing F_t (e.g., Revenue, Cash, or Operating Expenses) exhibit the identical 10^3 scaling deficit relative to their running baselines, apply the global integer multiplier:

$$x_t^* = x_t \times 10^3 \tag{8}$$

1.7.2 Cases 3 & 4: 10-K vs. 10-Q Intra-Period Share Discrepancies

Symptom:

- *Case 3:* Standalone Q4 basic shares derived via annual subtraction ($\text{Shares}_{\text{10-K}} - \sum_{i=1}^{3} \text{Shares}_{\text{Q}i}$) implies 500M, whereas the concurrent filing or subsequent comparative table states 600M (with $t-1$ and $t+1$ stable at 600M).
- *Case 4:* 10-K reported weighted-average shares equals 600M, but a raw quarterly extract reports 500M against flanking baselines of 600M.

Diagnosis: Conflation of two distinct statutory disclosures:

1. *Point-in-Time Balance Sheet Common Stock:* Instantaneous shares outstanding as of the balance sheet date.
2. *Income Statement Weighted-Average Shares:* Cumulative or quarterly time-weighted shares utilized for EPS calculation.

Annual 10-K filings disclose full-year weighted averages, which cannot be decomposed by naive differencing if share repurchases or issuances occurred non-linearly across quarters.

Resolution Rule: Never compute Q4 share counts via residual subtraction of full-year weighted averages. Enforce balance sheet snapshot shares outstanding from the document cover page or balance sheet face. If an isolated quarterly figure violates the flanking consensus ($x_{t-1} = x_{t+1} = 600\text{M}$) and the 10-K period-end cover count confirms 600M, override the residual with the primary point-in-time balance sheet count:

$$x_t^* = x_{\text{cover}}(t) \tag{9}$$

1.7.3 Case 5: Pathological Data Truncation and Parse Collapse

Symptom: $\text{Shares}_{t-1} \approx 400\text{M} \longrightarrow \text{Shares}_t = 1\text{K} \longrightarrow \text{Shares}_{t+1} \approx 400\text{M}$.

Diagnosis: A non-scale parsing failure (e.g., table column shift, par value extraction, or corrupted OCR/XBRL integer truncation to minimum values). The ratio does not map to a standard scale exponent:

$$\frac{x_{t-1}}{x_t} = \frac{400\text{M}}{1\text{K}} = 400,000 \notin \{10^2, 10^3, 10^6\} \tag{10}$$

Resolution Rule: Do not attempt numerical healing. Quarantine the record as defective, drop x_t, and forward-fill from the last verified Point-in-Time state:

$$x_t^* = x_{t-1} \quad \text{with metadata flag: } \texttt{DATA_CORRUPTION_QUARANTINED} \tag{11}$$

1.7.4 Case 6: ASC 805-50 Common-Control Restatement Leakage

Symptom: SATS Total Assets: $\text{Assets}_{2022Q3} \approx \$6.1\text{B} \longrightarrow \text{Assets}_{2022Q4} \approx \$58.7\text{B} \longrightarrow \text{Assets}_{2023Q1} \approx \$6.2\text{B}$.

Diagnosis: Under US GAAP ASC 805-50, mergers of entities under common control (e.g., EchoStar acquiring DISH Network) mandate retrospective restatement of all historical comparative columns. A vendor parser extracted the restated 2022 comparative balance sheet from the post-merger 2023 10-K and injected it into the historical 2022Q4 time slice, violating PIT neutrality.

Resolution Rule: Apply the bi-directional spike rejection filter:

$$\text{Flag if } \ln\left(\frac{x_t}{x_{t-1}}\right) > \theta \quad \text{AND} \quad \ln\left(\frac{x_t}{x_{t+1}}\right) > \theta \quad \text{AND} \quad \left|\ln\left(\frac{x_{t-1}}{x_{t+1}}\right)\right| \leq \delta \tag{12}$$

where $\theta = 1.5$ ($\approx 4.5\times$) and $\delta = 0.35$. Discard any record whose SEC acceptance timestamp $\tau_{\text{accept}} > t_{\text{sim}}$, and revert to the original as-reported filing ($\approx \$6.15\text{B}$).

1.7.5 Case 7: Regulatory Asset Expansion and Repeal (SAB 121 / SAB 122)

Symptom: COIN Total Assets: $\text{Assets}_{2022Q1} \approx \$21\text{B} \longrightarrow \text{Assets}_{2022Q2\ldots2024Q3} \in [\$105\text{B}, \$348\text{B}] \longrightarrow \text{Assets}_{2025Q1\ldots} \approx \$22\text{B}$.

Diagnosis: SEC Staff Accounting Bulletin No. 121 (SAB 121) mandated that digital asset custodians recognize customer safeguarding obligations as both an on-balance-sheet asset and a corresponding liability at fair value. When legislative or regulatory adjustments (e.g., SAB 122 relief) permitted de-recognition, the gross asset balance sheet deflated. This is neither a vendor bug nor a permanent M&A expansion, but an accounting pass-through distortion.

Resolution Rule: Do not scale. Monitor the equity-to-asset capitalization ratio:

$$\text{CapRatio}_t = \frac{\text{Total Stockholders' Equity}_t}{\text{Total Assets}_t} \tag{13}$$

During the custodial expansion, Stockholders' Equity remains constant ($\sim \$5.5\text{B} - \$8\text{B}$), causing CapRatio_t to plummet from $\approx 25\%$ to $< 3\%$. Standardize fundamental factors by deducting safeguarding liabilities:

$$\text{Operating Assets}_t = \text{Total Assets}_t - \text{Custodial Safeguarding Liabilities}_t \tag{14}$$

1.8 Summary of Ingestion Actions

Case ID	Phenomenon	Primary Signature	Engine Resolution Action
Case 1	Missing share scale factor	Exact $\Delta \ln \approx 6.91$, cross-confirmed	Scale up by 10^3 across filing
Case 2	Missing asset scale factor	Exact $\Delta \ln \approx 6.91$, cross-confirmed	Scale up by 10^3 across filing
Case 3	10-K implied subtraction deficit	Discrepancy against cover snapshot	Adopt period-end cover common count
Case 4	Quarterly weighted-average defect	Single-period interim divergence	Align to balance sheet snapshot count
Case 5	Unstructured parse truncation	Irregular ratio $\notin \{10^2, 10^3, 10^6\}$	Quarantine record; forward-fill x_{t-1}
Case 6	ASC 805 common-control restatement	Spike-revert needle ($\ln > 1.5$, revert < 0.35)	Strip restatement; enforce PIT acceptance date
Case 7	SAB 121/122 custodial mandate	Gross expansion, flat equity, matched liab.	Isolate Operating Assets = Assets − $\text{Liab}_{\text{custodial}}$

Table 2: Taxonomy of Fundamental Ingestion Failures and Systematic Resolution Policies.

2 Total Liabilities

Current Liabilities :
Current Debt (Short-Term Borrowings)
\+ Accounts Payable
\+ Accrued & Other Current Liabilities
=**Total Current Liabilities** (**TCL**)
+Long-Term Debt & Capital Lease Obligations
+Deferred Revenue (Non-Current / Other)
+Other Non-Current Liabilities
=**Total Liabilities** (**TL**)

Stockholders′ Equity :
Preferred Stock
\+ Common Stock
\+ Additional Paid-in Capital (APIC)
\+ Retained Earnings / (Deficit)
± Accumulated Other Comprehensive Income (AOCI)
− Treasury Stock
\+ Other Equity
=**Total Equity Attributable to Parent**
+Non-Controlling (Minority) Interest
=**Total Equity** (**TE**)

=**Total Liabilities** & **Equity** (**TL**&**E**)

Following the common-size balance sheet framework presented by BARUCH-FIN3000 [2019].

2.1 Total Current Liabilities

Current Liabilities :
Current Debt (Short-Term Borrowings)
\+ Accounts Payable
\+ Accrued & Other Current Liabilities
=**Total Current Liabilities** (**TCL**)

Following the common-size balance sheet framework presented by BARUCH-FIN3000 [2019].

Parallel to asset allocation, we deconstruct the right-hand side of the balance sheet into its primary funding sources: operating liabilities, debt financing, and shareholders' equity. Following standard corporate finance nomenclature [BARUCH-FIN3000, 2019]

2.2 Total Current Liabilities

Listing 4 details the audit query used to evaluate the internal consistency between reconstructed current liabilities and the reported aggregate.

```
WITH latest_balance_sheets AS (
    SELECT
        *,
        ROW_NUMBER() OVER (
            PARTITION BY ticker, period_end, timeframe
            ORDER BY filing_date DESC
```

```
        ) AS rn
    FROM balance_sheets
),
ratios as (
    select
        ticker,
        period_end,
        timeframe,
        COALESCE(debt_current , 0) as debt_current,
        COALESCE(accounts_payable , 0) as accounts_payable,
        COALESCE(accrued_and_other_current_liabilities , 0) as
    accrued_and_other_current_liabilities,
    COALESCE(deferred_revenue_current , 0) as deferred_revenue_current,
        COALESCE(total_current_liabilities , 0) as total_current_liabilities
    from latest_balance_sheets
  where timeframe = 'quarterly'
),
prep AS (
    SELECT
        ticker,
        period_end,
        total_current_liabilities,
        (
            COALESCE(debt_current, 0)
            + COALESCE(accounts_payable, 0)
            + COALESCE(accrued_and_other_current_liabilities, 0)
            + COALESCE(deferred_revenue_current, 0)
        ) AS r_tcl
    FROM ratios
    WHERE total_current_liabilities IS NOT NULL
      AND total_current_liabilities <> 0
),
evaluated AS (
    SELECT
        ticker,
        period_end,
        ROUND(r_tcl, 4) AS r_total_current_liabilities,
        ROUND(total_current_liabilities, 4) AS total_current_liabilities,
        1.0000 AS target_ratio,
        ROUND((r_tcl / total_current_liabilities) - 1.0, 4) AS ratio_discrepancy,
        ABS((r_tcl / total_current_liabilities) - 1.0) AS abs_discrepancy
    FROM prep
)
SELECT
    ticker,
    period_end,
    r_total_current_liabilities,
    total_current_liabilities,
    target_ratio,
    ratio_discrepancy
FROM evaluated
WHERE abs_discrepancy >=0.01
ORDER BY abs_discrepancy DESC;
```

Listing 4: Total Current Liability Reconstruction Discrepancy Query

Query | Query History

```
        ticker,
        period_end,
        ROUND(r_tcl, 4) AS r_total_current_liabilities,
        ROUND(total_current_liabilities, 4) AS total_current_liabilities,
        1.0000 AS target_ratio,
        ROUND((r_tcl / total_current_liabilities) - 1.0, 4) AS ratio_discrepancy,
        ABS((r_tcl / total_current_liabilities) - 1.0) AS abs_discrepancy
    FROM prep
)
SELECT
    ticker,
    period_end,
    r_total_current_liabilities,
    total_current_liabilities,
    target_ratio,
    ratio_discrepancy
FROM evaluated
WHERE abs_discrepancy >=0.01
ORDER BY abs_discrepancy DESC;
```

Data Output | Messages | Graph Visualiser | Notifications

	ticker character varying (20)	period_end date	r_total_current_liabilities numeric	total_current_liabilities numeric	target_ratio numeric	ratio_discrepancy numeric
1	VFF	2020-12-31	60209000.0000	72280000.0000	1.0000	-0.1670
2	PHM	2025-03-31	2521073000.0000	2983051000.0000	1.0000	-0.1549
3	PHM	2021-06-30	4655406000.0000	4770925000.0000	1.0000	-0.0242
4	KR	2017-01-28	12609000000.0000	12860000000.0000	1.0000	-0.0195
5	PBF	2016-03-31	1593032000.0000	1623003000.0000	1.0000	-0.0185
6	KR	2016-01-30	12750000000.0000	12971000000.0000	1.0000	-0.0170
7	KR	2016-05-21	12924000000.0000	13145000000.0000	1.0000	-0.0168
8	KR	2016-08-13	13300000000.0000	13521000000.0000	1.0000	-0.0163
9	KR	2016-11-05	13903000000.0000	14124000000.0000	1.0000	-0.0156
10	PBF	2016-09-30	2021301000.0000	2047733000.0000	1.0000	-0.0129
11	PBF	2016-06-30	1876988000.0000	1901518000.0000	1.0000	-0.0129
12	FSV	2016-12-31	173180000.0000	175122000.0000	1.0000	-0.0111

Total rows: 12 | Query complete 00:00:00.046

Figure 9: Current liability reconstruction discrepancies ($\geq 1\%$), affecting 12 of 9,439 quarterly observations.

Discrepancy Analysis

Across 9,439 quarterly filings evaluated, only 12 observations exhibited a component reconstruction discrepancy exceeding 1.0%. In each case, reported liabilities exceeded the reconstructed sum, indicating that omissions stem from industry-specific reporting lines rather than phantom liabilities:

- **PulteGroup (`PHM`) — Captive Finance & Unclassified Operations:** On 2025-03-31, reconstructed liabilities totaled \$2,521.07M versus reported \$2,983.05M (a gap of \$461.98M, or −15.5%). As a homebuilder operating a mortgage banking subsidiary (Pulte Mortgage), liabilities include warehouse credit facilities, customer escrow deposits, and obligations on consolidated inventory not owned. Because these captive finance obligations do not map to the primary four current liability columns, they fall out of the standard corporate sum.
- **Village Farms (`VFF`) — Current Operating Lease Liabilities (ASC 842):** On 2020-12-31, reconstructed liabilities totaled \$60.21M versus reported \$72.28M (a difference of \$12.07M, or −16.7%). Under ASC 842/IFRS 16, short-term lease obligations are categorized as `operating_lease_liability_current`. Because this column is absent from the standard four-column schema, capitalized lease payments create a structural deficit in agricultural and greenhouse operators with heavy leased infrastructure.
- **Kroger (`KR`) — Unmapped Accruals & Taxes Payable:** On 2017-01-28, reconstructed liabilities totaled \$12,609.00M versus reported \$12,860.00M (−1.9%). Retail grocery chains carry self-insurance reserves and income taxes payable due within one year. When vendor parsers do not bundle tax liabilities into `accrued_and_other_current_liabilities`, a mild deficit occurs.

2.3 Total Liabilities

Current Liabilities :
Current Debt (Short-Term Borrowings)
\+ Accounts Payable
\+ Accrued & Other Current Liabilities
=**Total Current Liabilities** (**TCL**)
+Long-Term Debt & Capital Lease Obligations
+Deferred Revenue (Non-Current / Other)
+Other Non-Current Liabilities
=**Total Liabilities** (**TL**)

Following the common-size balance sheet framework presented by BARUCH-FIN3000 [2019].

Listing 5 details the audit query used to evaluate the internal consistency of reconstructed liabilities against reported total liabilities across quarterly balance sheets.

```sql
WITH latest_balance_sheets AS (
    SELECT
        *,
        ROW_NUMBER() OVER (
            PARTITION BY ticker, period_end, timeframe
            ORDER BY filing_date DESC
        ) AS rn
    FROM balance_sheets
),
prep AS (
    SELECT
        ticker,
        period_end,
        total_liabilities,
        (
            COALESCE(total_current_liabilities, 0)
            + COALESCE(long_term_debt_and_capital_lease_obligations, 0)
            + COALESCE(other_noncurrent_liabilities, 0)
        ) AS r_tl
    FROM latest_balance_sheets
    WHERE
  timeframe = 'quarterly'
  AND rn = 1
    AND total_liabilities <> 0
    AND total_liabilities is NOT NULL
),
evaluated AS (
    SELECT
        ticker,
        period_end,
        ROUND(r_tl, 4) AS r_total_liabilities,
        ROUND(total_liabilities, 4) AS total_liabilities,
        1.0000 AS target_ratio,
        ROUND((r_tl / total_liabilities) - 1.0, 4) AS ratio_discrepancy,
        ABS((r_tl / total_liabilities) - 1.0) AS abs_discrepancy
    FROM prep
)
SELECT
    ticker,
    period_end,
    r_total_liabilities,
    total_liabilities,
    target_ratio,
    ratio_discrepancy
FROM evaluated
WHERE abs_discrepancy >=0.01
ORDER BY abs_discrepancy DESC;
```

Listing 5: Total Liability Reconstruction Discrepancy Query

Query | Query History

```
        ticker,
        period_end,
        ROUND(r_tl, 4) AS r_total_liabilities,
        ROUND(total_liabilities, 4) AS total_liabilities,
        1.0000 AS target_ratio,
        ROUND((r_tl / total_liabilities) - 1.0, 4) AS ratio_discrepancy,
        ABS((r_tl / total_liabilities) - 1.0) AS abs_discrepancy
    FROM prep
)
SELECT
    ticker,
    period_end,
    r_total_liabilities,
    total_liabilities,
    target_ratio,
    ratio_discrepancy
FROM evaluated
WHERE abs_discrepancy >=0.01
ORDER BY abs_discrepancy DESC;
```

Data Output | Messages | Graph Visualiser | Notifications

	ticker character varying (20)	period_end date	r_total_liabilities numeric	total_liabilities numeric	target_ratio numeric	ratio_discrepancy numeric
1	CBNA	2026-06-30	7438000.0000	2008907000.0000	1.0000	-0.9963
2	CBNA	2025-03-31	6963000.0000	1575355000.0000	1.0000	-0.9956
3	CBNA	2025-06-30	6264000.0000	1288179000.0000	1.0000	-0.9951
4	CBNA	2025-09-30	6719000.0000	1371259000.0000	1.0000	-0.9951
5	CBNA	2025-12-31	7900000.0000	1581180000.0000	1.0000	-0.9950
6	CBNA	2024-12-31	6943000.0000	1256878000.0000	1.0000	-0.9945
7	CBNA	2026-03-31	13065000.0000	1743799000.0000	1.0000	-0.9925
8	COSO	2026-03-31	28480000.0000	2085624000.0000	1.0000	-0.9863
9	CCB	2022-03-31	49360000.0000	2625830000.0000	1.0000	-0.9812
10	CCB	2022-06-30	54756000.0000	2752061000.0000	1.0000	-0.9801
11	FGBI	2021-12-31	57739000.0000	2654231000.0000	1.0000	-0.9782
12	MVBF	2026-06-30	69934000.0000	3201310000.0000	1.0000	-0.9782
13	CCB	2026-03-31	118903000.0000	5160067000.0000	1.0000	-0.9770

Total rows: 940 | Query complete 00:00:00.065

Figure 10: Total liability reconstruction discrepancies ($\geq 1\%$), affecting 940 of 9,617 quarterly observations ($\approx 9.77\%$).

Analysis: Financial Institutions and Unclassified Balance Sheets

The query flagged 940 discrepancies across 9,617 quarterly observations ($\approx 9.77\%$). The most extreme deviations cluster between -98% and -99%, dominated by regional banks and financial holding companies:

- `CBNA` — Chain Bridge Bancorp (Commercial Bank)
- `COSO` — Coastal Financial / Community Banking Entity
- `MVBF` — MVB Financial Corp (Commercial Banking)
- `FGBI` — First Guaranty Bancshares (Regional Bank)

The standard corporate reconstruction formula assumes:

$$\text{Reconstr. Liabil.} = \text{Total Current Liabil.} + \text{Long-Term Debt} + \text{Other Noncurrent Liabil.} \quad (15)$$

This formulation systematically fails for commercial banks due to GAAP accounting standards (ASC 942):

1. **Omission of Customer Deposits (85%–95% of Liabilities):** A bank's core liabilities consist of customer deposits (checking, savings, money market accounts, CDs), repo facilities, and Federal Home Loan Bank (FHLB) advances. These instruments are categorized neither as corporate current liabilities nor as funded long-term debt. Because the schema lacks a dedicated `deposits` column, the vast majority of the liability structure is entirely omitted.

2. **Absence of Classified Balance Sheets:** Financial institutions operate with unclassified balance sheets that do not segregate assets or obligations into "current" versus "noncurrent" maturity buckets. Consequently, `total_current_liabilities` is typically null or zero, leaving only incidental holding-company obligations (such as minor corporate notes) captured by the formula.

3 Total Equity

	Stockholders′ Equity :
	Preferred Stock
+	Common Stock
+	Additional Paid-in Capital (APIC)
+	Retained Earnings / (Deficit)
±	Accumulated Other Comprehensive Income (AOCI)
−	Treasury Stock
+	Other Equity
=	**Total Equity Attributable to Parent**
+	Non-Controlling (Minority) Interest
=	**Total Equity** (**TE**)

Following the common-size balance sheet framework presented by BARUCH-FIN3000 [2019].

```sql
WITH latest_balance_sheets AS (
    SELECT
        *,
        ROW_NUMBER() OVER (
            PARTITION BY ticker, period_end, timeframe
            ORDER BY filing_date DESC
        ) AS rn
    FROM balance_sheets
),
prep AS (
    SELECT
        ticker,
        period_end,
        total_equity,
        (
            COALESCE(preferred_stock, 0)
            + COALESCE(common_stock, 0)
            + COALESCE(additional_paid_in_capital, 0)
            + COALESCE(retained_earnings_deficit, 0)
            + COALESCE(accumulated_other_comprehensive_income, 0)
            + COALESCE(treasury_stock, 0)
            + COALESCE(other_equity, 0)
            + COALESCE(noncontrolling_interest, 0)
        ) AS r_te
    FROM latest_balance_sheets
    WHERE timeframe = 'quarterly'
      AND rn = 1
      AND total_equity <> 0
      AND total_equity IS NOT NULL
),
evaluated AS (
    SELECT
        ticker,
        period_end,
        ROUND(r_te, 4) AS r_total_equity,
        ROUND(total_equity, 4) AS total_equity,
        1.0000 AS target_ratio,
        ROUND((r_te / total_equity) - 1.0, 4) AS ratio_discrepancy,
        ABS((r_te / total_equity) - 1.0) AS abs_discrepancy
    FROM prep
)
SELECT
    ticker,
    period_end,
    r_total_equity,
    total_equity,
    target_ratio,
    ratio_discrepancy
```

```
FROM evaluated
WHERE abs_discrepancy >=0.00
ORDER BY abs_discrepancy DESC;
```

Listing 6: Total Equity Reconstruction Discrepancy Query

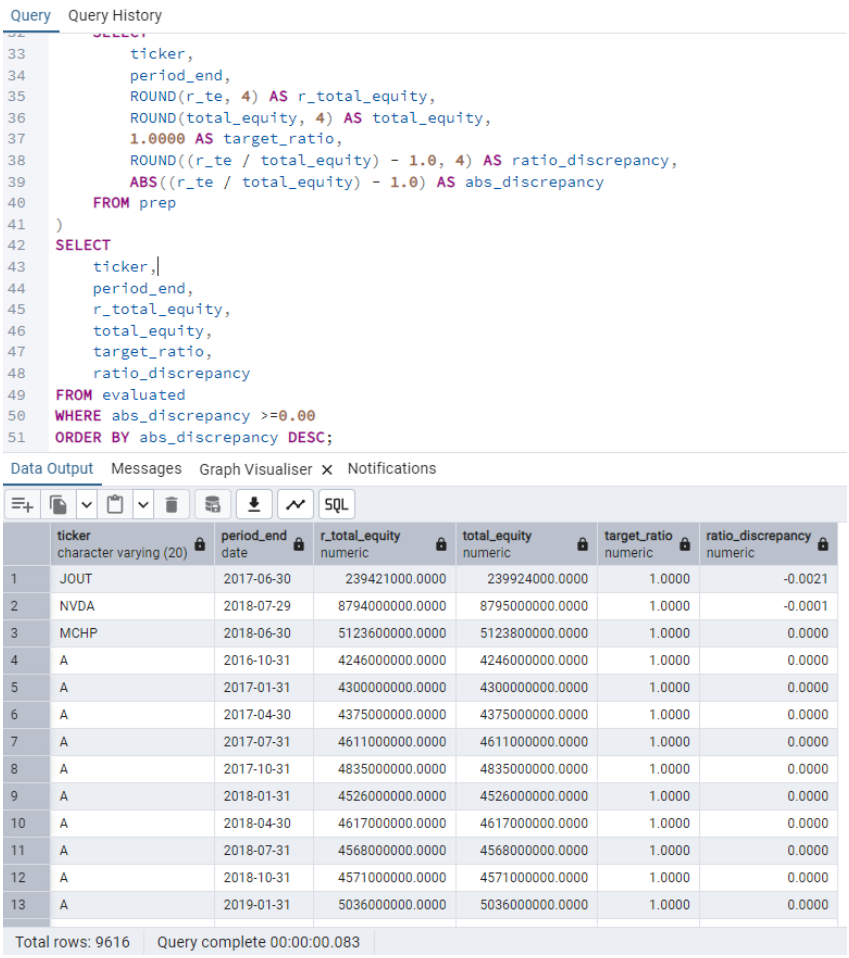


Figure 11: 0 rows affected, only 2 rounding error from 2017-2018

4 Total Total Asset, Liabilities, and Equity

Total Liabilities (**TL**)

+**Total Equity** (**TE**)

=**Total Liabilities** & **Equity** (**TL**&**E**)

Following the common-size balance sheet framework presented by BARUCH-FIN3000 [2019].

Listing 7 details the verification query used to test whether r_ta = Total Assets within a 1.0% relative tolerance across canonical quarterly filings.

```sql
WITH latest_balance_sheets AS (
    SELECT
        *,
        ROW_NUMBER() OVER (
            PARTITION BY ticker, period_end, timeframe
            ORDER BY filing_date DESC
        ) AS rn
    FROM balance_sheets
),
prep AS (
    SELECT
        ticker,
        period_end,
        total_assets,
        total_equity,
        total_liabilities,
        (
            COALESCE(total_equity, 0)
            + COALESCE(total_liabilities, 0)
        ) AS r_ta
    FROM latest_balance_sheets
    WHERE timeframe = 'quarterly'
      AND rn = 1
      AND total_assets IS NOT NULL
      AND total_assets <> 0
),
evaluated AS (
    SELECT
        ticker,
        period_end,
        ROUND(r_ta, 4) AS r_total_assets,
        ROUND(total_assets, 4) AS total_assets,
        1.0000 AS target_ratio,
        ROUND((r_ta / total_assets) - 1.0, 4) AS ratio_discrepancy,
        ABS((r_ta / total_assets) - 1.0) AS abs_discrepancy
    FROM prep
)
SELECT
    ticker,
    period_end,
    r_total_assets,
    total_assets,
    target_ratio,
    ratio_discrepancy
FROM evaluated
-- WHERE abs_discrepancy > 0.01
ORDER BY abs_discrepancy DESC;
```

Listing 7: Balance Sheet Identity Reconciliation Query ($A = L + E$)

Discrepancy Analysis: Mezzanine and Redeemable Equity

Across 9,617 deduplicated quarterly records, 417 observations ($\approx 4.34\%$) failed the identity $(TL + TE)/TA = 1.0 \pm 0.01$. Rather than signaling mathematical non-compliance by the reporting entities, these deviations isolate an architectural limitation of two-sided schemas when processing securities subject to conditional redemption:

Query Query History

```
        ticker,
        period_end,
        ROUND(r_ta, 4) AS r_total_assets,
        ROUND(total_assets, 4) AS total_assets,
        1.0000 AS target_ratio,
        ROUND((r_ta / total_assets) - 1.0, 4) AS ratio_discrepancy,
        ABS((r_ta / total_assets) - 1.0) AS abs_discrepancy
    FROM prep
)
SELECT
    ticker,
    period_end,
    r_total_assets,
    total_assets,
    target_ratio,
    ratio_discrepancy
FROM evaluated
WHERE abs_discrepancy > 0.01
ORDER BY abs_discrepancy DESC;
```

Data Output Messages Graph Visualiser Notifications

	ticker character varying (20)	period_end date	r_total_assets numeric	total_assets numeric	target_ratio numeric	ratio_discrepancy numeric
1	CNTX	2021-09-30	-18805319.0000	2259304.0000	1.0000	-9.3235
2	CVRX	2021-06-30	-274921000.0000	55062000.0000	1.0000	-5.9929
3	CCIV	2020-12-31	-1091395000.0000	1402681000.0000	1.0000	-1.7781
4	CCIV	2021-06-30	1966288.0000	2071966288.0000	1.0000	-0.9991
5	CCIV	2021-03-31	3181075.0000	2073181075.0000	1.0000	-0.9985
6	IRHO	2026-05-31	493985.0000	234030433.0000	1.0000	-0.9979
7	IRHO	2026-02-28	962821.0000	232424677.0000	1.0000	-0.9959
8	FCUV	2025-12-31	2939230.0000	8885514.0000	1.0000	-0.6692
9	STXS	2017-12-31	4706728.0000	10667203.0000	1.0000	-0.5588
10	MDXG	2023-03-31	75665000.0000	168159000.0000	1.0000	-0.5500
11	MDXG	2022-06-30	76173000.0000	168667000.0000	1.0000	-0.5484
12	MDXG	2022-12-31	78936000.0000	171430000.0000	1.0000	-0.5395
13	MDXG	2022-09-30	80304000.0000	172798000.0000	1.0000	-0.5353

Figure 12: Balance sheet identity discrepancies ($> 1\%$), affecting 417 of 9,617 quarterly observations ($\approx 4.34\%$).

- **Pre-Merger SPACs (`CCIV`, `IRHO`):** For Churchill Capital Corp IV (`CCIV`) on 2021-06-30, reported total assets equaled \$2,071.97M, whereas reconstructed assets totaled only \$1.97M (a ratio discrepancy of -0.9991, or -99.9%). Under SEC Rule 5-02.27 of Regulation S-X and ASC 480-10-S99, Class A shares subject to redemption must be classified outside of permanent stockholders' equity. Because SPAC trust proceeds reside in this "mezzanine" section, formulas summing only permanent liabilities and permanent equity omit $> 99\%$ of the capital structure.

- **Negative Reconstructed Assets in Early-Stage Biotechs (`CNTX`, `CVRX`):** Context Therapeutics (`CNTX`) reported assets of \$2.26M against a reconstructed total of $-\$18.81M$ (discrepancy of -9.3235). Similarly, CVRx (`CVRX`) showed \$55.06M in assets against $-\$274.92M$ in reconstructed assets. Prior to IPO, venture-backed biotechs issue convertible preferred stock with conditional redemption rights (categorized under mezzanine equity) to fund negative operating cash flows. The accumulated deficit in permanent retained earnings drives permanent stockholders' equity deeply negative. Omitting the positive mezzanine tranche leaves a negative equity balance that overwhelms liabilities, yielding a mathematically negative reconstructed asset total.

- **Post-DeSPAC Complex Capital Structures (`LCID`):** Lucid Group (`LCID`) on 2026-06-30 exhibited a -37.75% deficit (\$4,792.57M reconstructed vs. \$7,698.83M reported assets) driven by redeemable noncontrolling interests and convertible senior preferred shares issued post-business combination.

Systemic Resolution: To the Securities and Exchange Commission, these balance sheets balance to the penny under the tripartite standard:

$$\text{Total Assets} = \text{Total Liabilities} + \text{Temporary / Mezzanine Equity} + \text{Permanent Equity} \quad (16)$$

When third-party relational schemas flatten the right-hand side into only `total_liabilities` and permanent `total_equity`, mezzanine equity vanishes. Quantitative factor pipelines should either explicitly incorporate `temporary_equity` into Equation above or exclude SIC 6770 (Blank Check / SPAC entities) and pre-revenue clinical development issuers from standard industrial screens.

References

BARUCH-FIN3000. *Principles of Finance: FIN 3000 (Baruch College)*. McGraw-Hill Education, New York, NY, 2019. ISBN 9781260301182. Custom edition prepared for the Bert W. Wasserman Department of Economics and Finance, Zicklin School of Business, Baruch College.

Michael J. Cooper, Huseyin Gulen, and Michael J. Schill. Asset growth and the cross-section of stock returns. *The Journal of Finance*, 63(4):1609–1651, 2008. ISSN 00221082, 15406261. URL http://www.jstor.org/stable/25094485.